Linux File Permissions

Linux uses file permissions to control who can read, write, and execute a file or directory.

There are three types of permissions:

Permission	Symbol	Meaning	Numeric Value
Read	r	Read file contents	4
Write	w	Modify or delete file contents	2
Execute	x	Execute a file	1
No Permission	-	No access	0

Three Types of Users

Linux divides users into three categories:

User	Symbol	Meaning
Owner	u	The user who owns the file
Group	g	Users belonging to the file's group
Others	o	All other users
All	a	Owner, group, and others

Example:

Use the following command:

ls -l

output:

-rwxr-xr-- 1 sanjida sanjida 1024 file1.txt

```
-  rwx      r-x      r--
    |        |        |
  Owner   Group   Others
```

Part	Permission	
-	File type	Regular file
rwx	Owner	Read, write, execute
r-x	Group	Read and execute
r--	Others	Read only

chmod Command (Change File Permissions)

The chmod command is used to change file or directory permissions.

Syntax:

chmod permission filename

There are two ways to change permissions:

1. Symbolic (Character) Method
2. Numeric (Octal) Method

Symbolic (Character) Method

In this method, we use characters to assign or remove permissions.

Permission symbols:

- r = Read
- w = Write
- x = Execute

Operators:

Operator	Meaning
+	Add permission
-	Remove permission
=	Set exact permission

Examples:

Example 1: Give read permission to the owner

chmod u+r file.txt

Example 2: Remove write permission from the group

chmod g-w file.txt

Example 3: Give read and write permissions to the owner

chmod u+rw file.txt

Example 4: Remove all permissions from others

chmod o-rwx file.txt

Example 5: Set exact permissions for the owner

chmod u=rw file.txt

Octal (Numeric) Method

In this method, permissions are represented using numbers.

Permission	Value
Read (r)	4
Write (w)	2
Execute (x)	1
No permission (-)	0

To calculate the permission value, add the corresponding numbers.

Example:

- rwx = 4 + 2 + 1 = 7
- rw- = 4 + 2 + 0 = 6
- r-x = 4 + 0 + 1 = 5
- r-- = 4 + 0 + 0 = 4
- -wx = 0 + 2 + 1 = 3
- -w- = 0 + 2 + 0 = 2
- --x = 0 + 0 + 1 = 1

- --- = 0 + 0 + 0 = 0

Using chmod with Octal Values

Example:

chmod 755 file.txt

chown Command (Change File Owner)

The chown command is used to change the ownership of a file or directory.

Syntax:
chown new_owner filename

Example 1: Change the owner of a file

sudo chown rahim file.txt

Example 2: Change both owner and group

sudo chown rahim:staff file.txt

Example 3: Change only the group

sudo chown :staff file.txt

Example 4: Change ownership of multiple files

sudo chown rahim file1.txt file2.txt

Important Linux Commands for Permissions

Command	Purpose	Example
ls -l	View file permissions	ls -l file.txt
chmod	Change permissions	chmod 755 file.txt
chown	Change owner	chown rahim file.txt
chgrp	Change group	chgrp staff file.txt
whoami	Show current username	whoami
id	Show user and group information	id

Task

Convert Octal to Character

Convert the following octal values into character permissions.

1. 755
2. 644
3. 700
4. 760
5. 777
6. 640
7. 751
8. 600

Convert Character to Octal

Convert the following character permissions into octal values.

1. rwxr-xr-x
2. rw-r--r--
3. rwx------
4. rwxrw-r--
5. rw-rw-r--
6. rwxrwx---
7. r--r--r--
8. rwxr-x---

Write Linux Commands

1. Give read permission to others for notes.txt.
2. Remove write permission from the group of data.txt.
3. Give full permission to the owner and read-only permission to others for test.txt.
4. Change the owner of report.txt to karim.
5. Change both owner and group of project.txt to rahim and staff.
6. Set permission 750 for script.sh.
7. Remove all permissions from others for file.txt.